

# CHE 555 – Homework 1

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# 1 Full Solution

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REGRESSION ANALYSIS REPORT

CHE 555 – Homework 1
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PROBLEM 1
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SETUP
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Curve:  $y = 2x^2 + 3x + 1$ 
The distance from a point  $(x, y(x))$  to the origin is
 $D(x) = \text{sqrt}[x^2 + y(x)^2]$ 
Minimising  $D^2$  :
 $D^2(x) = x^2 + (2x^2 + 3x + 1)^2$ 
 $dD^2(x^*)/dx = 2x + 2(2x^2 + 3x + 1)(4x + 3) = 0$ 
 $2x + 2(2x^2 + 3x + 1)(4x + 3) = 0 \rightarrow 8x^3 + 22x^2 + 19x + 3 = 0$ 
The roots of this cubic equations were solved numerically using
↪ the Newton-Raphson method.

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Optimal x      = -0.36380427
 $y(x^*)$       =  $2(-0.363804)^2 + 3(-0.363804) + 1$ 
               = 0.17329428
 $D^2(x^*)$      =  $(-0.363804)^2 + (0.173294)^2$ 
               = 0.16238446
 $D(x^*)$       =  $\sqrt{0.16238446} = 0.40296955$ 

ANSWER
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Nearest point : (-0.363804, 0.173294)
Distance      : 0.402970

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PROBLEM 2
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Variables : x = side length of square base (cm)
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$h = \text{height (cm)}$

Volume constraint :  $x^2 \cdot h = 1000 \rightarrow h = 1000 / x^2$

Surface area (no top lid) :

$$\begin{aligned} S(x) &= x^2 + 4 \cdot x \cdot h \\ &= x^2 + 4 \cdot x \cdot (1000/x^2) \\ &= x^2 + 4000/x \end{aligned}$$

#### ANALYTICAL SOLUTION

$$dS/dx = 2x - 4000/x^2 = 0$$

$$2x^3 = 4000$$

$$x^3 = 2000$$

$$x^* = 2000^{(1/3)}$$

$$d^2S/dx^2 = 2 + 8000/x^3 > 0 \text{ for all } x > 0$$

#### NUMERICAL VALUES

$$x^* = 2000^{(1/3)} = 12.599210 \text{ cm}$$

$$h^* = 1000 / x^{*2} = 1000 / 12.599210^2 = 6.299605 \text{ cm}$$

$$S^* = (12.5992)^2 + 4000/12.5992 = 476.220316 \text{ cm}^2$$

$$d^2S/dx^2|_{x^*} = 6.0000 > 0$$

#### ANSWER

Optimal base side length : 12.5992 cm

Optimal height : 6.2996 cm

Minimum surface area : 476.2203 cm<sup>2</sup>

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↪

#### PROBLEM 3

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↪

Minimise:  $f(x_1, x_2) = 4x_1 - x_1^2 - 12$

Subject to:

$$g_1 : 25 - x_1^2 - x_2^2 = 0$$

$$g_2 : 10x_1 - x_1^2 + 10x_2 - x_2^2 - 34 \geq 0$$

$$g_3 : (x_1-3)^2 + (x_2-1)^2 \geq 0$$

$$x_1, x_2 \geq 0$$

VARIABLE COUNT

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Total Variables : 2 ( $x_1, x_2$ )

### DOF ANALYSIS

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Each independent equality constraint reduces DOF by 1.

Equality constraints : 1

DOF = total variables - equality constraints =  $2 - 1 = 1$

Number of independent variables : 1

One valid choice :  $\{ x_1 \}$

### EXPLANATION

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E1 defines  $x_2$  as a function of  $x_1$ :  $x_2 = \sqrt{(25 - x_1^2)}$  (with  $x_2 \geq 0$ ).  
 $\hookrightarrow 0$ ).

Once  $x_1$  is chosen (satisfying  $0 \leq x_1 \leq 5$ ),  $x_2$  is fully determined.

The inequality constraints further restrict the feasible range of  
 $\hookrightarrow x_1$

but do not reduce the number of independent variables.

$g_3 = (x_1 - 3)^2 + (x_2 - 1)^2 \geq 0$  is always satisfied (sum of squares).

### ANSWER

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Total variables : 2

Independent variables : 1

One possible choice :  $\{ x_1 \}$

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$\hookrightarrow$  **PROBLEM 4**

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$\hookrightarrow$  **GIVEN**

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Cost per barrel :  $C(P) = 50 + 0.1P + 9000/P$  (\$/barrel)

Selling price : \$300 / barrel

P = production rate (barrels/day)

#### PART A – Minimise cost per barrel

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$$dC/dP = 0.1 - 9000/P^2 = 0$$

$$P^2 = 9000 / 0.1 = 90000$$

$$P^* = \sqrt{90000} = 300 \text{ barrels/day}$$

$$d^2C/dP^2 = 18000/P^3 > 0 \text{ for } P > 0$$

Numerical check:

$$P^* = 300.0 \text{ barrels/day}$$

$$C(P^*) = 50 + 0.1(300.0) + 9000/300.0 = 110.00 \text{ \$/barrel}$$

$$d^2C/dP^2|P^* = 0.000667 > 0$$

PART B – Maximise daily profit

$$\text{Revenue per day : } R(P) = 300 \cdot P$$

$$\text{Total cost / day : } TC(P) = C(P) \cdot P = 50P + 0.1P^2 + 9000$$

$$\begin{aligned} \text{Profit / day : } \Pi(P) &= R - TC = 300P - 50P - 0.1P^2 - 9000 \\ &= 250P - 0.1P^2 - 9000 \end{aligned}$$

$$d\Pi/dP = 250 - 0.2P = 0$$

$$P^* = 250 / 0.2 = 1250 \text{ barrels/day}$$

$$d^2\Pi/dP^2 = -0.2 < 0$$

Numerical check:

$$P^* = 1250 \text{ barrels/day}$$

$$\text{Revenue} = 300 \times 1250 = \$375,000.00/\text{day}$$

$$\text{Total cost} = 50(1250) + 0.1(1250)^2 + 9000 = \$227,750.00/\text{day}$$

$$\text{Profit } \Pi(P^*) = \$375,000.00 - \$227,750.00 = \$147,250.00/\text{day}$$

ANSWERS

(A) Production minimising cost/barrel :  $P = 300$  barrel/day  $\rightarrow C =$   
 $\hookrightarrow \$110.00/\text{barrel}$

(B) Production maximising daily profit:  $P = 1250$  barrel/day  $\rightarrow \Pi =$   
 $\hookrightarrow \$147,250.00/\text{day}$

$\hookrightarrow$

PROBLEM 5

$\hookrightarrow$

DATA

| x  | y    |
|----|------|
| 10 | 1.00 |
| 20 | 1.26 |
| 30 | 1.86 |
| 40 | 3.31 |
| 50 | 7.08 |

## MODELS TO FIT

Model 1 :  $y = \exp(a + b \cdot x)$  [exp-linear]  
Model 2 :  $y = \exp(a + b \cdot x + c \cdot x^2)$  [exp-quadratic]  
Model 3 :  $y = a \cdot x^b$  [power law]

## METHOD

Models 1 & 2 are linearised by taking  $\ln y$  and fitting via OLS.  
Model 3 is linearised as  $\ln y = \ln a + b \cdot \ln x$  and fitted via OLS.  
Full regression statistics ( $R^2$ , adj- $R^2$ , SE, RMSE, F-test, t-tests, ANOVA table) are reported for each model in the sections below.

### Regression details:

↪ \_\_\_\_\_  
Model:  $y = \exp(a + b \cdot x)$

↪ \_\_\_\_\_  
Observations : 5  
Parameters (p) : 2  
DOF (error) : 3

— Goodness of Fit —  
 $R^2$  : 0.955527  
 $R^2$  (adjusted) : 0.940703  
SE of regression : 0.192228  
RMSE : 0.148900  
SSE : 0.110855  
SSR : 2.381815  
SST : 2.492670

— Overall F-Test —  
F-statistic : 64.4574  
F p-value : 0.00403481

— Parameter t-Tests ( $H_0$ : param = 0,  $\alpha = 0.05$ ) —  
Param Estimate Std Err t-stat p-value 95% CI  
↪ lower 95% CI upper

↪ \_\_\_\_\_  
a -0.662933 0.201611 -3.2882 0.046141  
↪ -1.304549 -0.021317 \*

b                    0.048804            0.006079            8.0285            0.00403481  
 ↳ 0.029458            0.068149            \*  
 (\* p < 0.05)

— ANOVA Table —

| Source       | df | SS       | MS       | F       |
|--------------|----|----------|----------|---------|
| ↳ p-value    |    |          |          |         |
| Regression   | 1  | 2.381815 | 2.381815 | 64.4574 |
| ↳ 0.00403481 |    |          |          |         |
| Residual     | 3  | 0.110855 | 0.036952 |         |
| Total        | 4  | 2.492670 |          |         |

↳ —  
 Model:  $y = \exp(a + b \cdot x + c \cdot x^2)$

↳ —  
 Observations : 5  
 Parameters (p) : 3  
 DOF (error) : 2

— Goodness of Fit —

R<sup>2</sup> : 0.999968  
 R<sup>2</sup> (adjusted) : 0.999936  
 SE of regression : 0.006314  
 RMSE : 0.003993  
 SSE : 0.000080  
 SSR : 2.492591  
 SST : 2.492670

— Overall F-Test —

F-statistic : 31265.9950  
 F p-value : 3.19826e-05

— Parameter t-Tests (H<sub>0</sub>: param = 0, α = 0.05) —

| Param   | Estimate     | Std Err | t-stat | p-value | 95% CI |
|---------|--------------|---------|--------|---------|--------|
| ↳ lower | 95% CI upper |         |        |         |        |

|   |           |           |         |             |  |
|---|-----------|-----------|---------|-------------|--|
| ↳ |           |           |         |             |  |
| a | -0.040266 | 0.013541  | -2.9736 | 0.096929    |  |
| ↳ | -0.098528 | 0.017997  |         |             |  |
| b | -0.004568 | 0.001032  | -4.4264 | 0.0474371   |  |
| ↳ | -0.009008 | -0.000128 | *       |             |  |
| c | 0.000890  | 0.000017  | 52.7166 | 0.000359642 |  |
| ↳ | 0.000817  | 0.000962  | *       |             |  |

(\* p < 0.05)

— ANOVA Table —

| Source        | df | SS       | MS       | F          |
|---------------|----|----------|----------|------------|
| ↪ p-value     |    |          |          |            |
| Regression    | 2  | 2.492591 | 1.246295 | 31265.9950 |
| ↪ 3.19826e-05 |    |          |          |            |
| Residual      | 2  | 0.000080 | 0.000040 |            |
| Total         | 4  | 2.492670 |          |            |

↪

Model:  $y = a \cdot x^b$  [log-log space]

↪

Observations : 5  
Parameters (p) : 2  
DOF (error) : 3

— Goodness of Fit —

$R^2$  : 0.818733  
 $R^2$  (adjusted) : 0.758311  
SE of regression : 0.388089  
RMSE : 0.300612  
SSE : 0.451838  
SSR : 2.040832  
SST : 2.492670

— Overall F-Test —

F-statistic : 13.5502  
F p-value : 0.0347317

— Parameter t-Tests ( $H_0$ : param = 0,  $\alpha$  = 0.05) —

| Param   | Estimate     | Std Err | t-stat | p-value | 95% CI |
|---------|--------------|---------|--------|---------|--------|
| ↪ lower | 95% CI upper |         |        |         |        |

|      |           |            |         |           |  |
|------|-----------|------------|---------|-----------|--|
| ln a | -2.863028 | 1.010440   | -2.8334 | 0.0660019 |  |
| ↪    | -6.078700 | 0.352644   |         |           |  |
| b    | 1.123962  | 0.305337   | 3.6811  | 0.0347317 |  |
| ↪    | 0.152245  | 2.095680 * |         |           |  |

(\* p < 0.05)

— ANOVA Table —

| Source    | df | SS | MS | F |
|-----------|----|----|----|---|
| ↪ p-value |    |    |    |   |



|             |   |          |          |         |
|-------------|---|----------|----------|---------|
| Regression  | 1 | 2.040832 | 2.040832 | 13.5502 |
| ↪ 0.0347317 |   |          |          |         |
| Residual    | 3 | 0.451838 | 0.150613 |         |
| Total       | 4 | 2.492670 |          |         |

Original-scale  $R^2$  for Model:  $y = a \cdot x^b$ : 0.726703 (adj  $R^2$ : 0.635604)

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#### MODEL COMPARISON SUMMARY

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| Model    |         | $R^2$ | adj $R^2$ | RMSE |
|----------|---------|-------|-----------|------|
| ↪ F-stat | F p-val |       |           |      |

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|               |             |          |          |          |
|---------------|-------------|----------|----------|----------|
| ↪             |             |          |          |          |
| exp_linear    |             | 0.955527 | 0.940703 | 0.148900 |
| ↪ 64.4574     | 0.00403481  |          |          |          |
| exp_quadratic |             | 0.999968 | 0.999936 | 0.003993 |
| ↪ 31265.9950  | 3.19826e-05 |          |          |          |
| power_law     |             | 0.726703 | 0.635604 | 0.300612 |
| ↪ 13.5502     | 0.0347317   |          |          |          |

Best model (highest adj  $R^2$ ): exp\_quadratic

#### FITTED VALUES vs. DATA

| x  | y_obs  | exp-lin | exp-quad | power  |
|----|--------|---------|----------|--------|
| 10 | 1.0000 | 0.8395  | 1.0030   | 0.7596 |
| 20 | 1.2600 | 1.3677  | 1.2513   | 1.6554 |
| 30 | 1.8600 | 2.2282  | 1.8650   | 2.6111 |
| 40 | 3.3100 | 3.6300  | 3.3210   | 3.6079 |
| 50 | 7.0800 | 5.9136  | 7.0651   | 4.6364 |

#### RESIDUALS

| x  | exp-lin | exp-quad | power   |
|----|---------|----------|---------|
| 10 | 0.1605  | -0.0030  | 0.2404  |
| 20 | -0.1077 | 0.0087   | -0.3954 |
| 30 | -0.3682 | -0.0050  | -0.7511 |
| 40 | -0.3200 | -0.0110  | -0.2979 |
| 50 | 1.1664  | 0.0149   | 2.4436  |

#### CONCLUSION

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The exp-quadratic model has the highest adjusted  $R^2$   
among the three candidates.  
Best model: exp-quadratic

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    END OF REPORT
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## 2 Plot

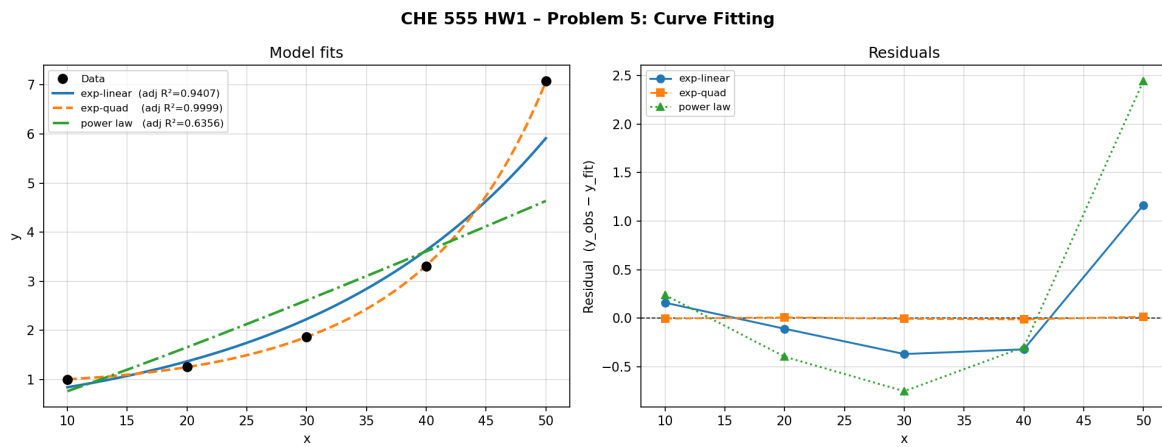


Figure 1: Curve fitting comparison for Problem 5.